Multiplication Table Generator

1. Problem Statement

Read a number from the user and print its multiplication table from 1 to 10 using a loop.

2. Objectives

1. Read a number from the user.
2. Use a `for` loop to repeat calculations.
3. Multiply the number by values from 1 to 10.
4. Display the multiplication table.

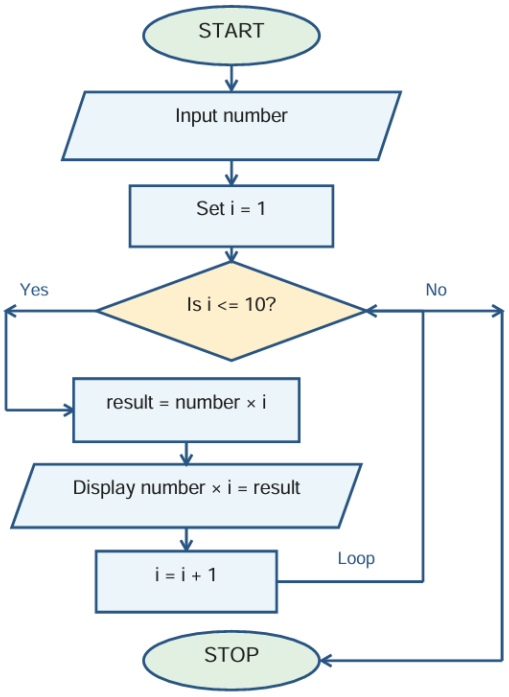
3. Assumptions

1. The user enters a valid integer.
2. The table is printed from 1 to 10.

4. Algorithm

1. Start.
2. Read a number.
3. Repeat `i` from 1 to 10.
4. Calculate `result = number × i`.
5. Display the multiplication statement.
6. Stop after the loop.

5. Flowchart



None

6. Pseudocode

None

```
INPUT number

FOR i = 1 TO 10
    result = number * i
    DISPLAY number, "x", i, "=", result
END FOR
```

8. Test Case and Expected Output

Input:

None

```
Enter a number: 5
```

Expected Output:

```
None
----- MULTIPLICATION TABLE -----
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

9. Conclusion

The program generates a multiplication table using a `for` loop. It repeats the calculation from 1 to 10 and displays the result for each multiplication.